

Chapter 32 — Embryology of the Human Genitourinary Tract

- **Two interrelated systems.** The urogenital system has a urinary part and a reproductive part, **most closely linked early in development.**
- **Timing.** Development **begins in the first trimester and is mostly complete by the end of the second trimester.**
- **The Carnegie collection** documents human embryos **from fertilisation to the indifferent stage (about 8–9 weeks).** After that, **androgens drive male and female structures apart.**
- **Why normal development matters.** It is the **basis for understanding congenital anomalies** and the first step in planning reconstruction.

DEVELOPMENT OF THE EXTERNAL GENITALIA

Genetic and hormonal control

- **The indifferent stage.** At 8–9 weeks the external genitalia consist of an **indifferent genital tubercle.**
- **The gonad comes first.** External genital differentiation **depends on prior gonadal differentiation.**
- **SRY.** Gonadal sex is set by the **SRY (sex-determining region Y) region on the short arm of the Y chromosome**, and its gene cascade directs testicular development.
- **Genes for the accessory structures** are carried on the **X chromosome and the autosomes.**
- **Leydig cells** appear at **about 8 weeks** and secrete testosterone.
- **Targets of testosterone and DHT.** Testosterone, and the more potent **dihydrotestosterone (DHT)**, masculinise:
 - the genital tubercle;
 - the Wolffian duct;
 - the **urogenital sinus (UGS).**
- **Type 2 5 α -reductase** converts testosterone to DHT. It is **expressed in the UGS and genital tubercle** and is **required for normal penile and prostatic development.**

Masculinisation and germ layers

- **Order of events: increased anogenital distance** (first sign) → **elongation of the genital tubercle** → **urethral plate** forms within it.
- **The genital tubercle** is the precursor of **both the penis and the clitoris** and contains all three germ layers.

GERM-LAYER ORIGIN OF THE GENITAL TUBERCLE

Germ layer	Derivative
Ectoderm	Skin of the phallus and the prepuce
Mesoderm	Corporal bodies (erectile tissue)
Endoderm	Urethral plate → penile urethra, reaching to near the distal tip of the future glans

- **The pattern across the rows:** one layer, one compartment. The **urethra is endodermal along its whole length**, which underpins the modern view of the glanular urethra.

Penile urethra — the "double zipper"

- **Opening zipper (canalisation).** The urethral plate **canalises from proximal to distal** → a **wide, diamond-shaped urethral groove** bounded by the urethral folds.
 - It is driven by **proliferation (Ki67-positive)**, with **no apoptosis** (caspase-3-negative).
 - Between **6.5 and 10.5 weeks** the meatus advances from the level of the scrotal folds to the proximal shaft. A distal **epithelial tag** of unknown significance is present in both sexes.
- **Closing zipper (fusion).** The **medial epithelial edges of the urethral folds fuse in the ventral midline** → the tubular penile urethra plus the **median raphe**.
 - A dense **vascular and neural network** (CD31- and PGP9.5-positive) lies at the site of fusion.
- **Epithelial markers:** **cytokeratin 6** marks the urethral plate and epidermis; **cytokeratin 7** marks urothelium.
- **Timetable:**
 - **11–12 weeks:** coronal sulcus;
 - **16 weeks:** folds fully fused along the shaft;
 - **18 weeks:** penile and urethral development essentially complete;
 - **20 weeks:** normal fetal ventral curvature (**chordee**) resolved.

SHAFT URETHRA VS GLANULAR URETHRA — TWO MECHANISMS

	Penile shaft urethra	Glanular urethra
Mechanism	Canalisation, then fusion of the urethral folds	Direct canalisation of the urethral plate , with epithelial remodelling inside the glans
Epithelial origin	Endoderm (UGS / urethral plate)	Endoderm (UGS / urethral plate), for most of its length
Clinical corollary	Failure of fusion → shaft hypospadias	The shaft–glans junction, where the mechanism changes, is the commonest site of the ectopic meatus

ORIGIN OF THE GLANULAR URETHRA — GLENISTER VS THE ENDODERMAL VIEW

	Ectodermal intrusion theory (Glenister) — outdated	Current (endodermal) view
Glanular urethra formed by	Skin (ectoderm) growing into the glans to meet the endodermal urethra	Canalisation of the endodermal urethral plate within the glans
Epithelium of the whole urethra	Mixed ectoderm and endoderm	Endoderm of the UGS throughout

- **Common ground:** on either view, **the meatus marks the ectoderm–endoderm (skin–urethral epithelium) junction**. At 18 weeks the distal urethra is still remodelling at this junction.

Prepuce

- **The prepuce depends on the urethra.** It forms at the same time as the glanular urethra and **depends on normal urethral development**.
- **Sequence:**
 1. At about 8 weeks, **preputial folds** appear on each side of the shaft.
 2. They join dorsally as a **flat preputial ridge** at the corona. Ventrally, the ridge is **blocked by the not-yet-formed glanular urethra**.
 3. The folds extend **distally** over the glans and **ventrally** to fuse in the midline.
 4. The epithelial **preputial lamina** separates the mesenchyme of the glans from that of the prepuce.
 5. **Postnatally, the lamina delaminates** → preputial space → the prepuce becomes retractable.

- **The prepuce is richly innervated.**
- **Proof of the urethral link.** In hypospadias, preputial tissue is **absent ventrally and excessive dorsally** (the dorsal hooded foreskin).

Penile innervation

- **Origin.** In **week 5**, neural crest cells form the peripheral and autonomic nervous systems.
- **Somatic supply. S2–S4 via the pudendal nerve**, which runs:
 - over the sacrospinous ligament;
 - through **Alcock's (pudendal) canal**;
 - through the transverse perineal muscle;
 - onto the dorsum as the **dorsal nerve of the penis**.
- **Autonomic supply.** From the **vesical and prostatic plexuses**:
 - **sympathetic L1–L2**;
 - **parasympathetic S2–S4**.
- **Cavernous nerves** leave the pelvis **between the transverse perineal muscles and the membranous urethra**, then pass beneath the pubic arch to each corpus cavernosum.
- **Dorsal nerve distribution**:
 - bundles at **11 and 1 o'clock**;
 - they fan around the tunica **as far as the junction of the corpora cavernosa with the corpus spongiosum**;
 - they leave a **nerve-free zone at 12 o'clock**.

Concept — *The penile urethra is built by two different mechanisms (fusion in the shaft, canalisation in the glans), and the prepuce depends on both. So the junction between the mechanisms predicts the commonest site of hypospadias, and failed ventral urethral closure predicts the hooded foreskin.*

Clinical Correlation: Hypospadias

- **Incidence:** about **1 in 250 newborn males**.
- **Definition** — **four components**:
 1. **ectopic urethral meatus**;
 2. **incomplete prepuce (dorsal hooded foreskin)**;
 3. **ventral skin deficiency**;
 4. possibly **penile curvature**.
- **Commonest meatal site:** the **junction of the shaft and the glans**, where fold fusion (shaft) gives way to direct canalisation (glans).
- **Anatomy.** The hypospadiac penis is **normal apart from the urethra, spongiosum, overlying skin and ventral prepuce**.
- **Arrest, not deformation.** Abortive urethral and spongiosal elements (epithelium, connective tissue, smooth muscle, vessels) lie in the adjacent tissue. So hypospadias is an **arrest of development, not a deformation**.
- **Aetiology**:
 - **unknown in the great majority** of cases;
 - working hypothesis: **genetic susceptibility plus maternal exposure to endocrine disruptors**.
- **A known genetic defect is rare.** Example: **type 2 5 α -reductase deficiency** → **severe hypospadias**. The enzyme is expressed at the fusion site and is needed to fuse the edges of the groove.

- **Chordee without hypospadias** can arise from **penoscrotal webbing**.

Clinical Correlation: Correction of Penile Curvature

- **Nerve map:**
 - along the shaft, nerves run on the tunica albuginea and **surround the corpora ventrally up to the spongiosal junction**;
 - from the glans to the separation of the crura (which attach to the pubic rami), they are **bundled at 11 and 1 o'clock**.
- **12 o'clock is the plication site for mild-to-moderate curvature**, because:
 - it is the **nerve-free zone**;
 - the **tunica albuginea is thickest** there, so the suture anchors well.
- **Scope:** curvature **with hypospadias** and **congenital curvature without it**.

Clinical Correlation: Duplicated Urethra

- **Rare**, in **hypospadiac** and **epispadiac** forms.
- **Embryology** — longitudinal splitting of the urethral plate:
 - **complete duplication:** the plate splits from the bladder distally;
 - **Y-duplication:** the plate splits along the shaft of the genital tubercle.
- **Signalling: Sonic hedgehog (Shh)** appears to control the process.

Concept — All three correlations apply the same developmental map. **Hypospadias is arrested development at the fusion–canalisation boundary, the safe zone for plication is the 12 o'clock nerve-free zone on the nerve map, and duplication is abnormal splitting of the single endodermal plate.**

DEVELOPMENT OF THE FEMALE EXTERNAL GENITALIA

- **The clitoris is built like the penis, minus the urethra.** It forms in the same way, **except that no urethra is incorporated**.
- **Vestibular plate.** It is the homologue of the urethral plate and **canalises (opening zipper)** into a wide, diamond-shaped **vestibular groove**.
- **Canalisation is androgen-independent**, because it happens in both sexes.
- **Fusion does not happen.** The **vestibular folds** (homologues of the urethral folds) **do not fuse** → an open vestibular groove. **Fusion is the androgen-dependent step.**
- **Glans clitoris.** It caps the narrowed paired corporal bodies. Where they fuse, they form a **midline septum** from the ventral side, running about **halfway into the glans**.
- **Innervation mirrors the penis:**
 - nerves are **densest dorsally**;
 - there is a **nerve-free zone at 12 o'clock**;
 - nerves extend ventrolaterally around the tunica;
 - the glans is supplied by **perforating dorsal nerve branches** at the dorsal corporoglandular junction;
 - glans nerve density is **lowest ventrally, beside the glans septum**.

MALE VS FEMALE EXTERNAL GENITALIA

Step	Male	Female
Plate	Urethral plate	Vestibular plate
Opening zipper (canalisation)	Yes	Yes — androgen-independent

Step	Male	Female
Closing zipper (fold fusion)	Yes → tubular penile urethra and raphe	No → open vestibular groove
Urethra within the organ	Penile urethra	None
Nerve-free zone	12 o'clock	12 o'clock
Orientation at 16 weeks	Penis at about 90° to the body	Clitoris recessed to the body wall

Clinical Correlation: Congenital Adrenal Hyperplasia

- **Congenital adrenal hyperplasia (CAH)** is **autosomal recessive** and **impairs cortisol synthesis** → **virilisation of 46,XX fetuses** and genital atypia at birth.
- **Differential diagnosis:** it can be mistaken for **hypospadias with bilateral non-palpable testes**.
- **Mechanism:** the vestibular groove **closes to a varying degree**, forming a **clitoral-shaft urethra anatomically indistinguishable from a male urethra**.
- **Most severe form:** a penis-like clitoris with a **terminal meatus, glans and foreskin**.
- **The only distinguishing features:** **no palpable gonads** and **excess pigmentation**.

Concept — *External genital sex comes down to **one androgen-dependent step, fusion of the folds**.*

Canalisation is common to both sexes. So androgen excess in a female (CAH) closes the vestibular groove into a male-type urethra, and androgen deficiency in a male leaves it open.

KEY POINTS: EXTERNAL GENITALIA

- **Androgens start differentiation.** The ambisexual genital tubercle begins to become a penis at **8 weeks**.
- **The penile urethra forms in two steps: opening zipper** (canalisation of the plate) → **closing zipper** (fusion of the folds).
- **Origin of the urethra.** The epithelium of the **entire male urethra is of urogenital sinus origin**.
- **The clitoris is analogous to the penis** but has **no urethra**.
- **The hypospadiac penis is otherwise normal.** Only the urethra, spongiosum, skin and ventral prepuce are abnormal.
- **12 o'clock nerve-free zone.** It is present on the penile and clitoral tunica and suits **dorsal plication** in mild-to-moderate curvature.

DEVELOPMENT OF THE HUMAN PROSTATE

- **First sign.** At **10 weeks**, prostatic buds emerge from the urogenital sinus epithelium (UGE) **caudal to the bladder base**.
 - Reports range from **9.5 to 10 weeks**, because fetal age is hard to estimate.
- **Evidence base.** Human fetal data are **sparse**. Mechanism and endocrinology come from **animal studies**:
 - **Jost** and **Price** established androgen dependence;
 - it was later confirmed in humans.
- **Six stages, the same in human and mouse:**
 1. prebud UGS;
 2. **solid epithelial buds** emerge from the UGE;
 3. elongation and branching;
 4. **canalisation**;
 5. **luminal and basal cell** differentiation;

6. secretory cytodifferentiation.

- **Endodermal origin.** The prostate develops **below the bladder**, just below the ureteric insertion, from **FOXA1-positive** (endodermal) UGE.
- **Endodermal UGE also gives:** bladder, urachus, urethra and bulbourethral glands.
- **Mesodermal Wolffian duct gives:** ureter, vas deferens and seminal vesicle. The **vas lies medial to the ureter** on the posterior bladder.

Verumontanum — a uniquely human feature

- **What it is:** a hillock on the **dorsal wall of the UGS**, elongated from cranial to caudal.
- **What opens on its apex:**
 - the **Wolffian ducts**;
 - the **fused Müllerian ducts (prostatic utricle)**.
- **Why it matters:** it is the **interface between mesodermal epithelium (Wolffian ducts, utricle) and endodermal UGE**.

Budding and branching

- **Site.** Most buds emerge in the **gutters lateral to the verumontanum**, where adult ducts open. **Mucosal glands arise from the ventral urethra.**
- **Budding is stochastic.** Cords of UGE grow in and then regress. **Fetal ventral buds are not represented in the adult gland.**
- **Branching.** From **about 11 weeks**, growth and branching follow a **fixed spatial pattern** that **sets up the adult zonal anatomy**.
 - Ductal length to the first branch point increases **from about 250 µm at 14 weeks to about 1000 µm at 17 weeks**.
- **Canalisation** runs **proximal (urethral) → distal (tips)**, together with **luminal and basal differentiation**.

TABLE 32.1 MARKER EXPRESSION PROFILES IN UROGENITAL SINUS EPITHELIUM (UGE), SOLID DUCTS, AND ADULT HUMAN PROSTATE

Marker group	Prebud UGE	Solid ducts	Adult luminal cells	Adult basal cells
Basal-type: TP63, KRT5, KRT14, KRT19, GSTpi	+	+	–	+
Luminal-type: KRT8, KRT18	+	+	+	–

Modified from Wang et al., 2001.

- **What the table shows:** precursor epithelium **co-expresses basal and luminal markers**.
- **The rule:** as cells differentiate, **future luminal cells keep luminal markers and lose basal ones; basal cells do the reverse**.
- **The trap:**
 - **TP63** and **KRT5/14** are **basal**; **KRT8/18** are **luminal**;
 - **KRT19** and **GSTpi** go with the **basal** set in the adult, though present in every precursor.

Stromal differentiation

- **What the stroma becomes.** Urogenital sinus mesenchyme (UGM) becomes **smooth muscle**, the main component of adult prostatic stroma.
- **Direction: ventral first, then lateral and dorsal:**
 - **9 weeks:** sparse ventrolateral cells;
 - **15 weeks:** ventral bundles;

- **19 weeks:** peripheral smooth muscle, where **branching is most intense near the capsule**.

Endocrinology

- **Androgens are essential.** Androgens from the **fetal testis** act through the **nuclear androgen receptor (AR)**, activated by testosterone or DHT.
- **Evidence for androgen dependence:**
 - there is **no prostate when the AR is non-functional** (mice and humans);
 - the **female UGS forms a prostate when exposed to androgen**.
- **DHT is critical.** Testosterone can bind the AR directly, but DHT:
 - is made **within the UGS by type 2 5 α -reductase (5 α R2)**, mainly in the **prostatic mesenchyme**;
 - has **about tenfold greater AR affinity** than testosterone.
- **Evidence for 5 α R2:**
 - **5 α R2 deficiency** → **rudimentary prostate** (and abnormal external genitalia);
 - mice lacking 5 α R1 and 5 α R2 have smaller prostates and seminal vesicles.
- **5 α R1's role is unclear.** The **5 α R1 gene is normal in 5 α R2-deficient patients**.

TESTOSTERONE VS DHT IN PROSTATIC DEVELOPMENT

	Testosterone	DHT
Source	Leydig cells of the fetal testis	Local conversion in the UGS mesenchyme by 5 α R2
AR affinity	Lower	About tenfold higher
Loss-of-function phenotype	—	5 α R2 deficiency → rudimentary prostate, severe hypospadias

Mesenchymal-epithelial interactions

- **Androgens act through the mesenchyme.** Androgen triggers the process through the AR, but **UGM induces and specifies prostatic identity in the epithelium**.
- **Instructive induction.** UGM can induce prostate even in **embryonic or adult bladder epithelium** (rodent and human).
- **Reciprocal.** Prostatic epithelium is needed for UGM to become smooth muscle.

Oestrogen effects

- **Oestrogens disrupt prostatic development.**
- **Acute effect: squamous metaplasia**, for example with **diethylstilbestrol (DES)**.
- **Receptor: ESR1 (oestrogen receptor α).**
 - Metaplasia occurs in wild-type and **ESR2 (oestrogen receptor β)-knockout** mice;
 - it does **not** occur in ESR1-null mice.

TABLE 32.2 SQUAMOUS METAPLASIA IN TISSUE RECOMBINANTS COMPOSED OF WILD-TYPE (WT) AND ESR1-NULL PROSTATIC EPITHELIAL AND MESENCHYME

Mesenchyme	Epithelium	Squamous metaplasia
Wild-type	Wild-type	+
Wild-type	ESR1-null	—
ESR1-null	Wild-type	—

Mesenchyme	Epithelium	Squamous metaplasia
ESR1-null	ESR1-null	–

- **What the table is for:** it shows which compartment needs the receptor. **Metaplasia occurs only with ESR1 in both stroma and epithelium.**
- **The trap:** thinking epithelial ESR1 is enough. **Losing it from either compartment abolishes the response.**
- **Clinical corollary.** Fetal prostatic squamous metaplasia is normal in the third trimester, because of **high maternal and placental oestrogen.**

Concept — *In prostatic development, the hormone acts through the mesenchyme and the mesenchyme instructs the epithelium.* The same two-compartment rule explains three findings: DHT is made in the stroma, stroma can impose prostatic identity even on bladder epithelium, and oestrogen-induced metaplasia needs receptor in both compartments.

Development of the Human Seminal Vesicle

- **Literature is minimal** — only two human papers.
- **Origin:** a diverticulum of the Wolffian duct at **10–12 weeks.**
- **Growth:** it elongates and folds back → a **hook-like duct**, with **extensive side ducts by months 4–5.**
- **Autonomic innervation** of the muscle coat is present by **13 weeks.**
- **Pathology:** malformation and malignancy are **extremely rare**; the main malformation is a **seminal vesicle cyst.**

PROSTATE VS SEMINAL VESICLE — ORIGIN

	Prostate	Seminal vesicle
Epithelial origin	Endodermal UGE (FOXA1-positive)	Mesodermal Wolffian duct
Onset	10 weeks (buds)	10–12 weeks (diverticulum)

KEY POINTS: PROSTATE DEVELOPMENT

- **Androgens from the fetal testis** are critical for prostatic development.
- **Mesenchymal–epithelial interactions** are fundamental, though the whole process is triggered by androgen acting through the AR.

DEVELOPMENT OF THE GONADS

The indifferent gonad

- **Common start.** Testes produce sperm and ovaries oocytes, but **development begins identically.**
- **Germinal ridge:** forms at **4 weeks** on the medial urogenital ridge, which contains the mesonephric duct and tubules.
- **Four components of the indifferent gonad:**
 1. **coelomic epithelium** of the germinal ridge;
 2. underlying **mesenchyme**;
 3. adjacent **mesonephros**;
 4. **primordial germ cells (PGCs).**
- **Shape tells the sex apart later.** The gonads are **macroscopically identical at 7–8 weeks.** After that, **testes become spherical**, while **ovaries stay oblong (and longer).**

Primordial germ cells and sex cords

- **Migration route:** PGCs arise in the **yolk sac wall in week 3** → **dorsal mesentery of the hindgut** → **gonadal ridges**.
- **Primary sex cords.** In **week 6**, surface epithelium invades the ridge mesenchyme to form them.
 - In humans they **do not form seminiferous cords**;
 - human **seminiferous cords appear to derive from mesonephric tubules**.
- **First sex-specific histology:**
 - **testis:** testicular cords at **about 6 weeks**;
 - **ovary:** at **about 8 weeks**, **segregation into a cortical zone (ovigerous cords) and a medullary zone**.
- **Origin of the ovarian zones — competing views:**
 - two successive proliferations of the germinal epithelium; or
 - ovigerous cords from mesonephric tubules.

Testicular Formation

- **SRY-dependent step.** Under the SRY gene product, medullary cells coalesce into **seminiferous cords of Sertoli cells enclosing PGCs**.
 - The testicular medulla is a zone of **mesenchymal-to-epithelial transition**.
- **Sertoli cells differentiate only if SRY protein is present.** Without it, the gonad becomes an ovary.
- **Sertoli-germ cell contact**, soon after PGC arrival, is thought to be **key to normal male gametes**.
- **Cords → tubules.** The cords **canalise into seminiferous tubules**.
- **Duct connections:**
 - the **rete testis** (thin-walled ducts) develops with the cords;
 - at the hilum it joins **5–12 residual mesonephric tubules**, which become the **efferent ductules** to the head of the epididymis;
 - the **full continuity** — tubules → rete → efferent ductules → epididymis — is **not established until the third month**.
- **Epididymis and vas deferens** come from the **Wolffian (mesonephric) duct**.
- **Shape and capsule.** The testis changes from oblong to round, **reducing its contact with the mesonephros**, and a thick **tunica albuginea** forms beneath the surface mesothelium.

Müllerian-inhibiting substance (MIS; anti-Müllerian hormone, AMH)

- **Source and action.** A glycoprotein from **Sertoli cells** that makes the **paramesonephric (Müllerian) ducts regress between weeks 8 and 10**.
- **Male Müllerian remnants:**
 - the **appendix testis** (a small epithelial cyst at the upper pole);
 - the **prostatic utricle** (a pouch at the apex of the verumontanum).
- **In females, MIS is absent**, so the Müllerian ducts persist and form **most of the female internal genitalia**.

Leydig cells and Wolffian derivatives

- **Leydig cells** differentiate from genital-ridge mesenchyme **from 8 weeks** in response to SRY, and **begin testosterone production at 8 weeks**.
- **Control of testosterone secretion:** **placental chorionic gonadotrophin early** → **fetal pituitary gonadotrophins later**.
- **Weeks 8–12:** testosterone turns the Wolffian ducts into the paired **epididymis, vas deferens and seminal vesicle**.

- **Remnants:**
 - the cranial Wolffian duct → **appendix epididymis**;
 - mesonephric tubules near the lower pole → **paradidymis**.

MIS VS TESTOSTERONE — THE TWO TESTICULAR HORMONES

	MIS (AMH)	Testosterone
Cell	Sertoli	Leydig
Target	Müllerian (paramesonephric) duct	Wolffian (mesonephric) duct, UGS, genital tubercle
Effect	Regression (weeks 8–10)	Stabilisation and differentiation (weeks 8–12) → epididymis, vas, seminal vesicle
Male remnants	Appendix testis, prostatic utricle	Appendix epididymis, paradidymis
If absent	Uterus and tubes persist	Wolffian ducts regress

Ovarian Formation

- **Why the ovary is the default. No Y chromosome → no SRY → no Sertoli-cell cords, no early MIS, no Leydig cells and no androgen, so female development ensues.**
- **First ovary-specific event: cortex and medulla segregate at about 8 weeks.** The cortex fills with **anastomosing cortical cords**.
- **Origin of the cortical cords is disputed:**
 - from mesonephric tubules; or
 - from gonadal-ridge epithelial-like cells at the surface.
 - Either way, the cords **break into clusters that germ cells enter**.
- **Second ovary-specific event: germ cell nests at 10–11 weeks.**
 - Nests are clusters of **2–4 germ cells joined by intercellular bridges**.
 - They are **clonal**, possibly from a single PGC.
 - They **enlarge by proliferation**, then **break up by germ-cell apoptosis**.

Folliculogenesis

1. **FOXL2-positive** somatic interstitial cells surround single germ cells → **primordial follicle**: oocyte plus a single layer of **squamous** cells.
2. **About 20 weeks**: the cells become **cuboidal** → **primary follicle**.
3. **Term**: granulosa cells proliferate → **secondary follicle**, with **multiple granulosa layers**.
4. **Postnatal**:
 - **antral follicles** with a fluid-filled cavity;
 - the **theca** organises around the follicle. It is vascular and:
 - supports the follicle;
 - **nourishes the avascular granulosa**;
 - **synthesises sex steroids**.
5. **Meiotic arrest**. Oogonia enlarge and **enter meiosis I as primary oocytes in fetal life**, then **arrest until puberty**. After that, oocytes resume one by one under the **monthly gonadotrophin surge**.

TESTIS VS OVARY — TRIGGER, TIMELINE AND DEFAULT

	Testis	Ovary
Trigger	SRY → Sertoli cells	Absence of SRY (default)
First specific feature	Testicular cords, about 6 weeks	Cortex and medulla, about 8 weeks
Second feature	Leydig cells and testosterone, 8 weeks	Germ cell nests, 10–11 weeks
Germ cells	Enclosed by Sertoli cells in cords → seminiferous tubules	Nests → primordial (squamous) → primary (cuboidal, ~20 weeks) → secondary (term) follicles
Shape	Spherical	Oblong, longer
Fetal hormones needed for duct development	MIS and testosterone	None

Concept — Gonadal sex is a **single switch, SRY in the pre-Sertoli cell**. Everything male follows from Sertoli cells: MIS and Leydig-cell androgen. Everything female follows from their absence. So the male phenotype is active and hormone-driven, while the female internal phenotype is the default.

TABLE 32.3 EMBRYOLOGIC ORIGIN OF UROGENITAL STRUCTURES

Embryonic structure	Male derivative	Female derivative
Indifferent gonad (cortex / medulla)	Testis; seminiferous tubules; rete testis (mediastinum testis)	Ovary; ovarian follicles; rete ovarii (hilum, analogous to rete testis)
Gubernaculum	Gubernaculum testis (descent into the scrotum)	Ovarian ligament; round ligament of the uterus
Mesonephric tubules (mesonephros, second kidney)	Efferent ductules; paradidymis	Epoöphoron, paroöphoron (non-functional, between the ovary and tube)
Wolffian (mesonephric) duct ; stalk of the ureteric bud	Appendix of epididymis; ductus deferens; ejaculatory duct and seminal vesicle ; ureter, pelvis, calyces, collecting tubules	Gartner duct (anterosuperior vaginal wall); ureter, pelvis, calyces, collecting tubules
Intermediate mesoderm (metanephric blastema)	Kidney (metanephros); nephrons (glomerulus, proximal tubule, loop of Henle, distal tubule)	Same
Müllerian (paramesonephric) duct	Appendix testis; prostatic utricle	Paratubal cyst ; uterine (fallopian) tube; uterus; cervix
Urogenital sinus	Bladder and trigone; urethra; prostate ; verumontanum; urethral glands (Littre); bulbourethral glands (Cowper)	Bladder and trigone; urethra; vagina, hymen ; urethral glands (Skene); greater vestibular glands (Bartholin)
Genital tubercle	Penis; glans; corpora cavernosa; corpus spongiosum	Clitoris; glans; corpora cavernosa; bulb of the vestibule
Urethral / vestibular folds	Ventral aspect of the penis	Labia minora
Labioscrotal swellings	Scrotum	Labia majora

Modified from Moore, Persaud and Torchia, 2016.

- **What the table is for:** a homologue map. Each primordium gives a paired male and female structure, so a remnant or anomaly in one sex can be predicted from the other.
- **Pattern across the rows:**
 - **Wolffian and mesonephric derivatives persist in the male and leave vestigial remnants in the female,** and the Müllerian duct shows the mirror image;

- **the ureter and collecting system come from the Wolffian system in both sexes**, which is why Wolffian anomalies carry renal anomalies (OHVIRA).
- **Homologue pairs:**
 - Littre ↔ Skene glands;
 - Cowper ↔ Bartholin glands;
 - corpus spongiosum ↔ bulb of the vestibule;
 - rete testis ↔ rete ovarii;
 - paradidymis ↔ paroöphoron.
- **The traps:**
 - the **vagina is listed under the UGS, not the Müllerian duct**;
 - the **appendix testis is Müllerian**, whereas the **appendix epididymis is Wolffian**.

Gonadal Descent

- **Starting position.** The urogenital ridge is **identical in both sexes at 7–8 weeks**. The gonad lies **near the kidney in the upper lumbar region**.
- **Two ligaments hold it:**
 - cranially, the **cranial suspensory ligament (CSL)**;
 - caudally, a ligament that becomes the **gubernaculum**.
- **Weeks 10–15:**
 - the **testis reaches the future inguinal region** as the abdomen enlarges;
 - the **ovary descends only into the pelvis**.
- **Male.** Androgen causes **CSL regression**, and the gubernaculum swells into a **plump ligament anchoring the testis at the groin** while the kidney moves cranially.
- **Female.** The **CSL persists**, keeping the ovary near the kidney; the **gubernaculum stays rudimentary**.

Transabdominal phase

- **Swelling reaction (gubernacular outgrowth).** The gubernaculum enlarges caudally.
- **Cord shortening.** The gubernacular cord is **taken up into the enlarging bulb**. This positions the testis over the inguinal ring so that **abdominal pressure can push it out**.
- **Cord transection misdirects the testis** → descent into the contralateral inguinal canal, or an aberrant intra-abdominal position.
- **Intra-abdominal pressure:** its role is **controversial here**, but it is **important in the inguinal and scrotal phases**.

Inguinoscrotal phase

- **Timing.** From the **seventh month**, the gubernaculum bulges beyond the external ring toward the scrotum.
- **Processus vaginalis.** At the same time the gubernaculum is **hollowed out by the processus vaginalis**, a peritoneal diverticulum that lets the testis leave the abdomen.
- **Distance.** The gubernaculum **migrates 3–5 cm**, and the processus lengthens with it.
- **Driving force:** intra-abdominal pressure, transmitted
 - **directly** via the lumen of the processus;
 - **indirectly** via the gubernacular cord.
- **Direction:** set by **androgen acting mostly indirectly, via the genitofemoral nerve**, which releases **calcitonin gene-related peptide (CGRP)**.

- **After descent:**
 - the **gubernacular bulb is resorbed**;
 - the **processus vaginalis closes, then involutes completely**.

THE TWO PHASES OF TESTICULAR DESCENT

	Transabdominal phase	Inguinoscrotal phase
Timing	Weeks 10–15 (ends at about 15 weeks)	From the 7th month
Key events	CSL regression ; gubernacular swelling; cord shortening	Gubernacular migration 3–5 cm to the scrotum; processus vaginalis lengthens; bulb regresses
Androgen dependence	CSL regression androgen-dependent ; gubernacular swelling androgen-independent (occurs in androgen resistance)	Androgen-dependent : absent in gonadotrophin deficiency and complete androgen resistance. Bulb regression androgen-dependent (the bulb stays enlarged in androgen resistance)
Other mediators	INSL3 (Leydig cell)	CGRP from the genitofemoral nerve
Intra-abdominal pressure	Controversial	Important
Failure →	Intra-abdominal testis	Inguinal or ectopic testis

INSL3

- **What it is: insulin-like peptide 3**, a Leydig-cell product identified in **1993**.
 - It is related to **relaxin and insulin**.
 - It is expressed in **fetal and adult Leydig cells**, depending on their differentiation.
- **Knockout phenotype: Insl3-null mice have intra-abdominal cryptorchidism with no other reproductive defect.**
- **Early orchidopexy restores fertility potential** in these mice.
- **Why it matters:** this is the **phenotype of classic human cryptorchidism**.
- **Androgen's overall role in descent is still unclear.** Defective androgen production or metabolism gives variable cryptorchidism.

Ovarian descent

- **A female gubernaculum exists.** It runs from the lower pole of the ovary to the labioscrotal folds, later passing through a formed inguinal canal.
- **It does not shorten**, but it **anchors the ovary in the pelvis → descent in the third month**, onto the **broad ligament**.
- **Week 7:** the gubernaculum attaches to the Müllerian ducts. As those ducts fuse, it is **tethered to the uterus** and divided into:
 1. **ovary → uterus: the ovarian ligament**;
 2. **uterus → inguinal canal → labia majora: the round ligament**.
- **Processus vaginalis in females:** normally obliterated. **If it stays patent → indirect inguinal hernia**.

Concept — Descent is a **tug between two ligaments followed by a migration**. Androgen removes the cranial restraint (CSL), INSL3 builds the caudal anchor (gubernaculum), and androgen via CGRP steers the inguinoscrotal migration. So the site of an undescended testis tells you which step failed.

Clinical Correlation: Cryptorchidism

- **Incidence:** about **3% of newborn males**. Hydrocele and hernia are more common still.

FAILED STEP OF DESCENT → ANOMALY

Failed step	Result
CSL retained	Intra-abdominal testis
Gubernacular attachment abnormal	Testis in the inguinal canal or just outside it
Gubernacular attachment ectopic	Ectopic testis: perineum, femoral region, or near the penis
Step 1 — gubernacular migration to the scrotum	Congenital cryptorchidism
Step 2 — closure of the processus vaginalis	Hydrocele or inguinal hernia
Step 3 — complete involution of the processus vaginalis	Acquired cryptorchidism (ascending testis): a fibrous remnant stops the cord lengthening after birth

Clinical Correlation: Streak Gonad

- **Three genotypes:**
 1. 45,X (Turner syndrome);
 2. 46,XX gonadal dysgenesis;
 3. 46,XY gonadal dysgenesis.
- **Mechanism** — both testicular hormones are absent:
 - no testosterone → **Wolffian ducts regress;**
 - no MIS → **Müllerian ducts form the tubes, uterus, cervix and upper vagina.**
- **Phenotype:** female-like internal and external genitalia.
- **Treatment:**
 - **hormone replacement** for secondary sexual characteristics, because the streak is non-functional;
 - **46,XY: risk of gonadoblastoma** and later malignancy → **streak gonads are typically removed.**

Clinical Correlation: Ovo-testicular Syndrome

- **Definition.** An ovotestis is a gonad containing **both testicular and ovarian tissue**. It is **extremely rare**.
- **Function.** Both tissues can function hormonally, before and after birth.
 - **Female fertility with normal oocytes is possible;**
 - **functioning adult spermatogenesis has not been reported.**
- **Ducts follow the gonad on the same side:**
 - **Müllerian structures** on a side **without testicular tissue;**
 - **Wolffian structures** on a side **with functioning testicular tissue.**
- **Genitalia:** often **atypical**.

Clinical Correlation: Persistent Müllerian Duct Syndrome

- **Definition.** A 46,XY male with a **uterus and fallopian tubes (hernia uteri inguinale)**.
- **Usual cause:** an **abnormal MIS receptor** → no MIS action → female internal genitalia with a **relatively normal male phenotype**.
- **Typical presentation:** found at **exploration for bilateral abdominal testes**.
- **Alternative cause:** an **immature testis** whose MIS production is **out of phase** with the Müllerian ducts' sensitivity, normally **9–13 weeks**.

MIS — TIMING OF REGRESSION VS WINDOW OF SENSITIVITY

Measure	Weeks	Meaning
Müllerian regression under MIS	8–10	When regression is observed
Müllerian sensitivity to MIS	9–13	Window in which the ducts can respond ; relevant to mistimed MIS

DISORDERS OF GONADAL AND DUCTAL DEVELOPMENT

Condition	Karyotype	Gonad	MIS	Testosterone	Internal ducts	External genitalia	Key point
Streak gonad	45,X / 46,XX / 46,XY	Streak, non- functional	Absent	Absent	Müllerian retained , Wolffian regressed	Female- like	Hormone replacement; remove the streak if 46,XY
Ovotesticular	—	Ovotestis	On the testicular side	On the testicular side	Follow the ipsilateral gonad	Often atypical	Female fertility possible; no spermatogenesis reported
Persistent Müllerian duct syndrome (PMDS)	46,XY	Testes (often abdominal)	Ineffective (receptor) or mistimed	Normal	Müllerian retained , Wolffian present	Male	Found at exploration for bilateral abdominal testes
CAH	46,XX	Ovaries	Absent	Excess (adrenal)	Female	Virilised ; can be male-identical	Impalpable gonads and pigmentation

- The organising pattern:
 - MIS decides the Müllerian ducts;
 - testosterone decides the Wolffian ducts and external genitalia;
 - each condition is a **different combination of the two signals**.

Concept — Every correlation in this section follows from **two independent testicular signals acting on two independent duct systems**. Remove both (streak gonad), block one (PMDS), give them unilaterally (ovotestis), or add androgen without MIS (CAH), and the phenotype can be predicted.

KEY POINTS: GONADAL DEVELOPMENT

- **SRY** makes the indifferent gonad of the ambisexual embryo **differentiate into a testis**.
- **Leydig-cell androgens** drive male genital differentiation, **including the prostate**.
- **Sertoli-cell MIS** makes the Müllerian structures (uterus and fallopian tubes) **degenerate**.
- **Both gonads reach the pelvis by the third month**. The testis enters the scrotum at about the seventh month, guided by the gubernaculum.

MÜLLERIAN STRUCTURES (FEMALE INTERNAL GENITALIA)

- **Why males lack them:** MIS from the testis.
- **Why females keep them.** Females have **no testosterone**, so the **Wolffian ducts resorb**. The Müllerian ducts persist and form the **uterine body, cervix and fallopian (uterine) tubes**.
- **In whole mounts the uterine body, cervix and vagina cannot be told apart**. The uterus enlarges with gestational age.

Development of the Human Female Reproductive Tract

Müllerian duct formation

1. **Start of week 5:** coelomic epithelium **invaginates on the lateral urogenital ridge**. These infoldings become the **tubal ostia**. The Müllerian duct lies **lateral to the mesonephros and Wolffian duct**.
2. The paired ducts **grow caudally using the Wolffian ducts as guidewires**. The **Wolffian duct is essential** for Müllerian migration to the UGS.
3. The contact between the ducts changes along their length:
 - cranially, **mesenchyme** separates them;
 - lower down, their **basement membranes** touch;
 - at the tip, their **epithelia** touch.
 - Cell exchange between the ducts is **not supported by lineage tracing**.
4. **Weeks 7–8:** the Müllerian ducts lie **between the two Wolffian ducts** as they approach the UGS.
 - At the indifferent stage (**about 54 days, Carnegie stage 22**) the **mesonephros is regressing from cranial to caudal**.
 - The point of contact with the UGS is the **Müllerian tubercle**, a term of limited value.
5. **Week 8:** the ducts **fuse in the midline → the uterovaginal canal**, with a **midline epithelial septum**.
6. **Week 9:** the **septum disappears**.
7. **The female Wolffian ducts degenerate**. Their remnants are:
 - **epoöphoron and paroöphoron** in the broad ligament;
 - **Gartner's duct** in the vaginal wall.

Uterine tube

- **Origin:** the paired **cranial, unfused** Müllerian ducts. **Fimbriae** come from the irregular ostia.
- **At 8 weeks** the tube is narrow, with a circular profile.
- **Later changes:**
 - the **mucosa becomes highly folded**, especially in the **infundibulum and ampulla**;
 - the mesenchyme divides into an **inner stroma** and an **outer smooth muscle layer**.

Uterine corpus and cervix

- **Origin:** the **cranial uterovaginal canal**. The caudal canal contributes to the vagina.
- **Septum.** The midline septum marks the fusion zone and is normally eliminated. **If it fails to regress → malformations.**
- **Corpus epithelium** is **simple columnar** and forms the **uterine and cervical glands**.
- **Cervical epithelium:**
 - **endocervix:** columnar and glandular;
 - near the **external os** it becomes **stratified squamous**, which also covers the **exocervix**.
- **Mesenchyme → endometrial stroma**, surrounded by the **myometrium**.

Vagina

- **The debate:** proposed contributions from the UGS, Müllerian ducts or Wolffian ducts, alone or in combination.

ORIGIN OF THE VAGINAL EPITHELIUM — KOFF VS BULMER VS IMMUNOHISTOCHEMISTRY

	Koff (1933)	Bulmer (1957)	Immunohistochemistry
Proposal	Müllerian and UGE (sinovaginal bulbs), with Müllerian predominance	UGE upgrowth forms the entire epithelial lining	Supports Bulmer
Method	Haematoxylin and eosin (H&E) only	H&E only	PAX2 (Müllerian) vs FOXA1 (endodermal UGE)

Sequence of vaginal development (the PAX2/FOXA1 evidence)

- Weeks 8–9:** the Müllerian ducts and uterovaginal canal are **PAX2-positive**.
- The caudal canal proliferates into the **solid vaginal plate**, which abuts the **FOXA1-positive sinovaginal bulbs** (UGS-derived).
- 12 weeks:** most of the plate is **PAX2-positive**; **FOXA1** is confined to the caudal plate, urethra and bladder.
- Weeks 14–21:** **FOXA1-positive UGE** progressively replaces **PAX2-positive Müllerian epithelium**.
- Canalisation:** starts at **about 15 weeks**, mostly complete by **18–19 weeks**.
 - The fornices are defined and the **exocervix projects into the vault**.
 - FOXA1** has reached **mid-vagina**.
- 21 weeks:**
 - the vaginal epithelium is **multilayered and glycogenated** (endogenous oestrogen);
 - FOXA1 runs from the introitus to the cervix**;
 - PAX2 covers the tube, uterus and cervix** (cervical epithelium stratified but thin);
 - the markers switch **abruptly at the vaginal–cervical junction**.
- Conclusions:**
 - the entire human vaginal epithelium is UGE-derived**;
 - the **hymen** is **FOXA1-positive**, so presumed **UGE-derived**.

UPPER VAGINA — TRANSIENT MÜLLERIAN ORIGIN, FINAL UGE LINING

Statement	What it refers to
Müllerian ducts give rise to the "upper vagina"	Classic teaching: the Müllerian (uterovaginal canal) origin of the upper vaginal plate
Entire vaginal epithelium is UGE-derived	The final epithelial lining , after UGE replaces the Müllerian epithelium up to the cervix

HUMAN VS MOUSE VAGINA

	Human	Mouse
Final vaginal epithelium	UGE (FOXA1) , replacing Müllerian epithelium up to the cervix	Müllerian (PAX2) , replacing the endodermal sinus vagina postnatally
Terminology	—	"Müllerian vagina" (cranial), "sinus vagina" (caudal)
Direction of replacement	Sinus replaces Müllerian	Müllerian replaces sinus

Concept — *The female tract is built by Müllerian ducts that depend on the Wolffian duct to reach the UGS, then fuse, then lose their septum. Each step has a failure mode, and the vagina is the exception whose lining is ultimately replaced by the sinus.*

Clinical Correlation: Vaginal Agenesis and Imperforate Hymen

- **Transverse vaginal septum:**
 - caused by **failure of the vaginal canal to canalise** → atresia or obstruction;
 - about **1 in 80,000 females**;
 - typically at the **junction of the middle and upper thirds**.
- **Imperforate hymen:**
 - caused by **failure of the inferior end to open**;
 - it is the **commonest cause of vaginal obstruction**;
 - it can present as a **bulging hymen with cyclic pain**, with a dilated vagina and uterus on ultrasound.

Clinical Correlation: Uterus Didelphys

- **Uterus didelphys:** failure of fusion of the caudal Müllerian ducts.
- **Unicornuate uterus:** one Müllerian duct fails to develop.

MÜLLERIAN ANOMALIES — FAILED STEP → ANOMALY

Failed step	Anomaly
One duct fails to develop	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys
Septum fails to regress	Uterine obstruction or duplication
Vaginal plate fails to canalise	Transverse vaginal septum (middle/upper third junction)
Inferior end fails to open	Imperforate hymen (commonest obstruction)
Müllerian and Wolffian ducts both abnormal	OHVIRA

Clinical Correlation: Obstructed Hemi-Vagina and Unilateral Renal Anomalies

- **Obstructed hemi-vagina and ipsilateral renal anomalies (OHVIRA):** a rare combined Müllerian and urinary tract anomaly, from abnormal Müllerian and Wolffian duct development.
- **Presentation:**
 - usually **after menarche**, with **cyclic abdominal pain and vaginal discharge**;
 - sometimes as an **acute abdomen or tubo-ovarian abscess**.
- **The renal anomaly is not always agenesis.** Reported variants include **ipsilateral dysplastic kidneys and ectopic ureters**, for example an ectopic ureter from a pelvic dysplastic nubbin draining into the obstructed hemi-vagina.
- **Why the kidney is often not seen on imaging:** usually **dysplasia and atrophy** from abnormal (not absent) Wolffian duct formation — **not true agenesis**.

TRUE RENAL AGENESIS VS OHVIRA RENAL ANOMALY

	True renal agenesis	OHVIRA renal anomaly
Cause	Absent Wolffian duct, ureteric bud or metanephric blastema	Abnormal Wolffian duct with abnormal ipsilateral Müllerian duct
Kidney	Absent	Dysplastic or atrophic nubbin , may drain via an ectopic ureter
Associated finding	Absent ipsilateral hemi-trigone	Ectopic ureter may enter the obstructed hemi-vagina

- **Why the distinction matters.** A dysplastic kidney with an ectopic ureter causes **persistent vaginal leakage after resection of the vaginal septum**. So **do not assume agenesis**.

- **Terminology:** OHVIRA's "RA" now stands for "**ipsilateral renal anomalies**".

KEY POINTS: MÜLLERIAN DEVELOPMENT

- **Vaginal epithelium.** Immunohistochemistry supports **Bulmer**: the entire human vaginal epithelium is **UGE-derived**, with no Müllerian or Wolffian cell origin.
- **The midline septum** marks the fusion zone and is normally eliminated. **Failure → uterine obstruction or duplication.**
- **Imperforate hymen** is the **commonest cause of vaginal obstruction.**

BLADDER: URETERAL DEVELOPMENT

- **Shared wall structure of bladder and ureter:**
 - urothelium-lined lumen;
 - **lamina propria** (extracellular matrix with disorganised smooth muscle);
 - outer **thick smooth muscle bundles**;
 - serosa.
- **Sex differences are small.** The ureters and bladder are alike in both sexes.
 - The exception: in males, **the prostate develops below the ureteric insertion**, so the **prostatic urethra is larger** than its female homologue.
- **The ureterovesical junction inserts into the trigone identically in both sexes.**

Formation of Urogenital Sinus

- **Division of the cloaca.** At weeks 5–6, the **urorectal septum** divides the **cloaca** into:
 - the **dorsal anorectal canal**;
 - the **ventral UGS**.
 - This follows the classic observations of Rathke and Tourneux.
- **Anorectal malformations** are thought to arise from the **cloacal membrane, not the urorectal septum.**
- **UGS and allantois.** Together they form the bladder and urethra. The **allantois is the precursor of the urachal remnant**; early on, the bladder is continuous with it.
- **The Wolffian ducts join the cloaca** just before it divides. **Their entry point is the landmark** between:
 - the **cephalad vesicourethral canal → bladder and pelvic urethra**;
 - the **caudal UGS → phallic urethra in males, and urethra and vaginal vestibule in females.**

Formation of Trigone

- **Weeks 5–6:** the **common excretory ducts** (the mesonephric ducts distal to the ureteric bud origin) **dilate and join the UGS.**
- **Apoptosis** lets the **ureters disconnect from the mesonephric ducts and fuse with the bladder.**

ORIGIN OF THE TRIGONE — CLASSIC VS CURRENT VIEW

	Classic hypothesis	Current (mouse lineage tracing)
Fibromuscular wall of the trigone	Extension of ureteral tissue into the bladder	Mostly bladder smooth muscle ; only a minor ureteral contribution

- **Orifice migration.** The ureteral orifices are incorporated into the bladder and **migrate cranially and laterally** within its base.

Weigert–Meyer rule (inferred mainly from duplex kidneys)

- **Rule:** the **upper-pole orifice rotates dorsally** relative to the lower-pole orifice and lies **more caudal and medial**.

WEIGERT–MEYER RULE — LOWER-POLE VS UPPER-POLE URETER

	Lower-pole ureter	Upper-pole ureter
Bud origin when abnormal	Too low on the mesonephric duct	Too high on the mesonephric duct
Incorporation	Premature , with excess migration	Delayed or incomplete
Orifice position	Lateral	Caudal and medial (ectopic)
Consequence	Vesicoureteral reflux: inadequate intramural tunnel	Ectopic insertion
Ectopic sites, male	—	Bladder neck, verumontanum, or Wolffian derivatives such as the vas deferens
Ectopic sites, female	—	Wolffian remnants (e.g. Gartner duct cyst) or the vaginal vestibule

- **Ectopic vas deferens.** It results from anomalous Wolffian development. The **vas may join the ureter** instead of the verumontanum, so that both drain into the prostatic urethra together.

Concept — *Where the ureteric bud arises on the Wolffian duct fixes where its orifice ends up. **Too low** → lateral orifice → reflux; **too high** → caudal and medial orifice → ectopia into Wolffian territory.*